

Boss Challenge — Paper 46 (Class 7)

Light | Motion & Time | Lines & Angles | Comparing Quantities

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Light always travels from one point to another along a path that is:
(A) a curved arc
(B) a widening spiral
(C) a straight line
(D) a wavy zig-zag
2. The distance an object covers in one unit of time is called its:
(A) speed
(B) force
(C) mass
(D) weight
3. When the measures of two angles together make exactly 90° , the angles are called:
(A) complementary angles
(B) adjacent angles
(C) supplementary angles
(D) vertically opposite angles
4. To write the fraction $\frac{3}{5}$ as a percentage, you compute:
(A) 53%
(B) 60%
(C) 35%
(D) 30%
5. When a ray of light strikes a polished mirror and bounces back, the bouncing-back of light is called:
(A) dispersion
(B) refraction
(C) absorption
(D) reflection
6. The SI (standard) unit of speed is:
(A) metre per second (m/s)
(B) second (s)
(C) metre (m)
(D) kilometre per hour (km/h)
7. When the measures of two angles together make exactly 180° , the angles are called:
(A) complementary angles
(B) supplementary angles
(C) right angles
(D) corresponding angles

8. Find the value of 25% of 80:

- (A) 25
- (B) 40
- (C) 20
- (D) 55

9. In the law of reflection, the angle of incidence is always measured between the incoming ray and the:

- (A) mirror surface itself
- (B) normal (line perpendicular to the mirror)
- (C) ground below
- (D) reflected ray

10. An object that covers equal distances in equal intervals of time is said to be in:

- (A) no motion (rest)
- (B) uniform motion
- (C) non-uniform motion
- (D) circular motion

11. An angle measures 35° . Its complement works out to:

- (A) 65°
- (B) 145°
- (C) 35°
- (D) 55°

12. The ratio 12 : 18 written in its simplest form is:

- (A) 1 : 2
- (B) 3 : 2
- (C) 2 : 3
- (D) 6 : 9

13. A ray of light hits a plane mirror at an angle of incidence of 35° . The angle of reflection is:

- (A) 70°
- (B) 35°
- (C) 55°
- (D) 0°

14. A car travels 150 km in 3 hours at a steady speed. Its speed is:

- (A) 45 km/h
- (B) 153 km/h
- (C) 450 km/h
- (D) 50 km/h

15. An angle measures 110° . Its supplement works out to:

- (A) 250°
- (B) 70°
- (C) 110°
- (D) 20°

16. A shopkeeper buys a toy for ₹200 and sells it for ₹250. His profit percent is:

- (A) 25%
- (B) 20%
- (C) 50%
- (D) 12.5%

17. The image of your face formed by a flat (plane) mirror is:

- (A) real and the same size
- (B) real, inverted and smaller
- (C) virtual, erect and the same size as your face
- (D) virtual, upside-down and larger

18. The instrument in a vehicle that measures its speed at that moment is the:

- (A) odometer
- (B) thermometer
- (C) barometer
- (D) speedometer

19. Where two straight lines intersect, the two angles lying directly across from each other at the point of crossing are:

- (A) equal (vertically opposite angles)
- (B) always complementary (add to 90°)
- (C) always 90° each
- (D) always supplementary (add to 180°)

20. A ₹500 jacket is sold at a 10% discount. The discount amount is:

- (A) ₹10
- (B) ₹450
- (C) ₹100
- (D) ₹50

21. When you raise your right hand in front of a plane mirror, the image appears to raise its left hand. This left-right swap is called:

- (A) lateral inversion
- (B) magnification
- (C) dispersion
- (D) refraction

22. A simple pendulum's one complete to-and-fro swing (there and back) is called one:
- (A) oscillation
 - (B) frequency
 - (C) time period for many swings
 - (D) amplitude
23. When two angles have the same vertex and one shared arm, sitting on opposite sides of that arm, they are called:
- (A) complementary angles
 - (B) vertically opposite angles
 - (C) alternate angles
 - (D) adjacent angles
24. Convert 0.75 into a percentage:
- (A) 0.75%
 - (B) 750%
 - (C) 7.5%
 - (D) 75%
25. A mirror that caves inward, the way the inside of a steel spoon's bowl does, is a:
- (A) plane mirror
 - (B) cylindrical mirror
 - (C) concave mirror
 - (D) convex mirror
26. The time taken by a simple pendulum to complete one oscillation is called its:
- (A) frequency
 - (B) amplitude
 - (C) speed
 - (D) time period
27. Two adjacent angles whose outer arms form a straight line make a:
- (A) complementary angles
 - (B) a right angle pair (90° total)
 - (C) linear pair (they add to 180°)
 - (D) vertically opposite angles
28. If 5 pens cost ■40, then by the unitary method 8 pens cost:
- (A) ■320
 - (B) ■50
 - (C) ■48
 - (D) ■64

- 29.** A convex mirror is used as a vehicle's side-view mirror mainly because it:
- (A) removes the driver's blind spot completely
 - (B) forms a real image on the road
 - (C) gives a wider field of view (covers more area behind)
 - (D) magnifies vehicles so they look closer
- 30.** On a distance-time graph, a straight line sloping upward represents:
- (A) uniform speed
 - (B) the object moving backward
 - (C) the object is at rest
 - (D) ever-increasing speed
- 31.** An angle that is greater than 90° but less than 180° is called:
- (A) a right angle
 - (B) an acute angle
 - (C) a reflex angle
 - (D) an obtuse angle
- 32.** A car's speed is 40 km/h in town and 60 km/h on the highway. By what percent is the highway speed greater than the town speed?
- (A) 60%
 - (B) 33%
 - (C) 50%
 - (D) 20%
- 33.** White sunlight passed through a glass prism spreads out into a band of seven colours. This splitting of light is called:
- (A) reflection
 - (B) dispersion
 - (C) lateral inversion
 - (D) absorption
- 34.** On a distance-time graph, a horizontal (flat) line means the object is:
- (A) moving at top speed
 - (B) moving backward
 - (C) speeding up
 - (D) at rest (not moving)
- 35.** A ray of light reflects off a mirror so that the angle of incidence equals the angle of reflection. If each is 50° , the angle between the incident and reflected rays is:
- (A) 40°
 - (B) 100°
 - (C) 130°
 - (D) 50°

36. In a class of 40 students, 60% passed a test. The number of students who passed is:

- (A) 24
- (B) 60
- (C) 30
- (D) 16

37. The correct order of the seven colours of the spectrum, from the one bent least to the one bent most by a prism, is remembered as:

- (A) RoyGBiv with red bent most
- (B) only three colours: red, blue, green
- (C) VIBGYOR (violet bends most, red least)
- (D) black, white and grey

38. To change a speed from km/h into m/s, you multiply by:

- (A) 1000
- (B) 5/18
- (C) 18/5
- (D) 3600

39. When a transversal cuts two parallel lines, a pair of corresponding angles are:

- (A) always 90°
- (B) complementary (add to 90°)
- (C) equal
- (D) supplementary (add to 180°)

40. Find the simple interest earned when ■1000 is kept at 5% per year for 2 years:

- (A) ■200
- (B) ■100
- (C) ■1100
- (D) ■50

41. A spoon's shiny outer (back) surface acts like a convex mirror. Looking into it, your image appears:

- (A) small and erect
- (B) the same size as a plane mirror
- (C) large and upside-down
- (D) real and projected onto the wall

42. A bus covers 40 km in the first hour and 60 km in the second hour. Its average speed over the 2 hours is:

- (A) 20 km/h
- (B) 50 km/h
- (C) 100 km/h
- (D) 10 km/h

43. When a transversal cuts two parallel lines, a pair of co-interior (allied) angles are:
- (A) complementary (add to 90°)
 - (B) equal
 - (C) supplementary (add to 180°)
 - (D) always 45° each
44. Out of 50 light bulbs in a shop, 40 are working. The percentage of working bulbs is:
- (A) 90%
 - (B) 80%
 - (C) 40%
 - (D) 20%
45. A torch beam in a dark room shows up as a sharp straight streak. This is direct evidence that light:
- (A) travels in straight lines
 - (B) only travels in water
 - (C) speeds up in a dark room
 - (D) bends around the dust
46. In the SI system of measurement, time is measured in the base unit called the:
- (A) hour
 - (B) minute
 - (C) day
 - (D) second
47. An angle that is exactly 180° looks like a straight line and is called a:
- (A) straight angle
 - (B) a right angle
 - (C) a complete angle
 - (D) a reflex angle
48. Two numbers are in the ratio 3 : 4 and their sum is 56. The smaller number is:
- (A) 32
 - (B) 24
 - (C) 28
 - (D) 8
49. Two plane mirrors are joined to make a kaleidoscope so that a few coloured beads form many beautiful symmetric patterns. The patterns are produced by:
- (A) dispersion of white light by the beads
 - (B) the beads multiplying inside the tube
 - (C) repeated reflections between the mirrors
 - (D) the glass bending the light by refraction

50. A car's speed rises from 40 km/h to 50 km/h. The percentage increase in its speed is:
- (A) 25%
 - (B) 10%
 - (C) 20%
 - (D) 50%
51. An angle whose measure lies somewhere between 180° and 360° is called:
- (A) a reflex angle
 - (B) a straight angle
 - (C) a complete angle
 - (D) an obtuse angle
52. A price rises from ₹80 to ₹100. The percentage increase is:
- (A) 25 rupees
 - (B) 125%
 - (C) 25%
 - (D) 20%
53. An object stands 8 cm in front of a plane mirror. The image is formed:
- (A) 4 cm behind the mirror
 - (B) 16 cm behind the mirror
 - (C) 8 cm behind the mirror
 - (D) right on the mirror surface
54. Two cars have speeds 60 km/h and 90 km/h. The ratio of the first car's speed to the second is:
- (A) 1 : 2
 - (B) 6 : 9 only
 - (C) 3 : 2
 - (D) 2 : 3
55. At exactly 3 o'clock, the angle made between a clock's hour hand and its minute hand is:
- (A) 90°
 - (B) 45°
 - (C) 180°
 - (D) 120°
56. Express 2 : 5 as a percentage:
- (A) 25%
 - (B) 20%
 - (C) 40%
 - (D) 50%
57. A magnifying glass that makes tiny print look larger is which kind of lens?
- (A) a plane glass sheet
 - (B) a concave (diverging) lens
 - (C) a convex mirror
 - (D) a convex (converging) lens

58. A train travels at a uniform 20 m/s. The distance it covers in 1 minute is:
- (A) 72000 m
 - (B) 120 m
 - (C) 20 m
 - (D) 1200 m
59. Through how many degrees does the minute hand of a clock turn in 15 minutes?
- (A) 45°
 - (B) 15°
 - (C) 180°
 - (D) 90°
60. A shirt costing ₦600 is sold at a loss of 15%. The loss amount is:
- (A) ₦510
 - (B) ₦90 profit
 - (C) ₦90
 - (D) ₦15
61. A ray of light strikes a plane mirror along the normal itself (straight on, angle of incidence 0°). It is reflected:
- (A) at 90° to the original path
 - (B) straight back along the same path
 - (C) at 45° to the mirror
 - (D) not reflected at all
62. Which of these natural events repeats at the most regular, fixed interval - useful as a way to measure time?
- (A) a falling leaf
 - (B) a clap of thunder
 - (C) a bursting balloon
 - (D) the swinging of a pendulum
63. Two complementary angles are in the ratio 2 : 3. The smaller angle is:
- (A) 54°
 - (B) 18°
 - (C) 60°
 - (D) 36°
64. A pendulum completes 45 swings out of an expected 60 in a faulty test. The percent of expected swings it managed is:
- (A) 60%
 - (B) 25%
 - (C) 75%
 - (D) 45%

65. A periscope, used to see over a wall, sends light around the bend using:
- (A) two plane mirrors set at 45°
 - (B) a single curved lens
 - (C) a convex mirror at the top only
 - (D) a glass prism that disperses light
66. The odometer of a vehicle records the:
- (A) fuel left in the tank
 - (B) engine temperature
 - (C) speed at this moment
 - (D) total distance the vehicle has travelled
67. Two supplementary angles are equal. Each angle measures:
- (A) 60°
 - (B) 45°
 - (C) 90°
 - (D) 180°
68. If 30% of a number is 60, the number is:
- (A) 200
 - (B) 600
 - (C) 18
 - (D) 90
69. A ray of light makes an angle of 20° with the surface of a plane mirror. Measured from the normal, its angle of incidence is:
- (A) 20°
 - (B) 45°
 - (C) 70°
 - (D) 90°
70. A boy walks to school at 4 km/h and cycles back along the same road at 12 km/h. Compared with walking, cycling is:
- (A) the same speed
 - (B) 8 times as fast
 - (C) 2 times as fast
 - (D) 3 times as fast
71. When a transversal cuts two parallel lines, a pair of alternate interior angles are:
- (A) supplementary (add to 180°)
 - (B) complementary (add to 90°)
 - (C) equal
 - (D) always 30° each

72. The ratio of 50 paise to ₹2 (in the same unit) is:

- (A) 25 : 1
- (B) 50 : 2
- (C) 1 : 2
- (D) 1 : 4

73. After the 20° -to-the-surface ray above reflects, the total angle between the incident ray and the reflected ray ($i + r$ from the normal) is:

- (A) 40°
- (B) 140°
- (C) 180°
- (D) 70°

74. Non-uniform motion is best described as motion in which the object:

- (A) always moves in a circle
- (B) covers equal distances in equal times
- (C) covers unequal distances in equal time intervals
- (D) never moves at all

75. The angle between the two hands of a clock at 6 o'clock is:

- (A) 60°
- (B) 360°
- (C) 180°
- (D) 90°

76. A bag has 20 red and 30 blue marbles. Red marbles form what percent of the total?

- (A) 66%
- (B) 60%
- (C) 20%
- (D) 40%

77. Which of these objects is luminous - it gives out its own light?

- (A) the Moon
- (B) the Sun
- (C) a mirror
- (D) a white wall

78. A 100-metre runner finishes in 20 seconds. Her average speed is:

- (A) 5 m/s
- (B) 80 m/s
- (C) 2000 m/s
- (D) 20 m/s

79. If one angle of a linear pair is 65° , the other angle is:
- (A) 25°
 - (B) 180°
 - (C) 65°
 - (D) 115°
80. A cyclist covers 36 km in 90 minutes. Her speed in km per hour is:
- (A) 24 km/h
 - (B) 36 km/h
 - (C) 40 km/h
 - (D) 54 km/h
81. What makes a real image different from a virtual one is that a real image:
- (A) is made only by plane mirrors
 - (B) is always upright
 - (C) appears behind the mirror
 - (D) can be caught (projected) on a screen
82. Light travels far faster than sound. In a thunderstorm you usually:
- (A) hear the thunder but never see lightning
 - (B) see the lightning before you hear the thunder
 - (C) hear the thunder before the flash
 - (D) see and hear them at exactly the same instant
83. Two lines in a plane that never meet, however far they are extended, are said to be:
- (A) intersecting
 - (B) perpendicular
 - (C) concurrent
 - (D) parallel
84. A number is increased by 20% and the result is 120. The original number was:
- (A) 140
 - (B) 24
 - (C) 100
 - (D) 96
85. A concave mirror is used in a torch or a car headlight behind the bulb because it can:
- (A) spread light in all directions
 - (B) send out a strong parallel beam of light
 - (C) split the light into seven colours
 - (D) shrink the bulb's image like a side-view mirror

86. A pendulum makes 30 complete oscillations in 60 seconds. Its time period (time for one oscillation) is:

- (A) 60 seconds
- (B) 30 seconds
- (C) 2 seconds
- (D) 0.5 seconds

87. Two lines that cross each other and make an angle of 90° at the crossing are called:

- (A) alternate lines
- (B) perpendicular lines
- (C) oblique lines
- (D) parallel lines

88. Half of a right angle, expressed as a percentage of a full turn (360°), is:

- (A) 50%
- (B) 45%
- (C) 12.5%
- (D) 25%

89. Sunlight reflected off calm water can dazzle your eyes. Compared with a rough, choppy water surface, the calm surface gives:

- (A) a clearer, mirror-like (regular) reflection
- (B) a spread-out, scattered glow
- (C) no reflection at all
- (D) colours split like a rainbow

90. Two buses leave together. Bus P goes 80 km in 2 h; bus Q goes 90 km in 3 h. The faster bus is:

- (A) bus Q, because it covered more distance
- (B) bus P (40 km/h vs 30 km/h)
- (C) both equal, since both are buses
- (D) bus Q, because it took the longest time

91. An angle of 360° (a full turn back to the starting arm) is called:

- (A) a straight angle
- (B) a complete angle
- (C) a reflex angle
- (D) a right angle

92. A 2-litre bottle is 35% full of juice. The amount of juice, in millilitres, is:

- (A) 700 mL
- (B) 35 mL
- (C) 70 mL
- (D) 1300 mL

93. Light from the Moon that lets us see it at night is:

- (A) the Moon's own glowing surface
- (B) sunlight reflected by the Moon
- (C) light made by the Moon's craters
- (D) starlight stored during the day

94. Distance is to the y-axis as time is to the x-axis on a distance-time graph. Choosing time along the bottom lets us read off:

- (A) how far the object has gone by any given time
- (B) the mass of the object
- (C) the temperature at each second
- (D) the colour of the moving object

95. The measure of an angle that is its own complement is:

- (A) 45°
- (B) 30°
- (C) 60°
- (D) 90°

96. A train's speed drops from 100 km/h to 75 km/h. The percentage decrease is:

- (A) 75%
- (B) 25%
- (C) 50%
- (D) 33%

97. Hold this printed word in front of a plane mirror: which letter looks exactly the same as its mirror image?

- (A) P
- (B) O
- (C) S
- (D) R

98. A car runs at 60 km/h. The time it takes to cover 150 km is:

- (A) 2.5 hours
- (B) 90 hours
- (C) 0.4 hours
- (D) 210 hours

99. Two angles forming a linear pair are in the ratio 1 : 2. The larger angle is:

- (A) 120°
- (B) 90°
- (C) 45°
- (D) 60°

100. A glass pane lets through 80% of the light falling on it. If 500 units of light strike it, the amount passing through is:

- (A) 80 units
- (B) 400 units
- (C) 580 units
- (D) 100 units